

Explaining the Variation in REIT Capital Structure:

The Role of Asset Liquidation Value

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Abstract

We test the Shleifer-Vishny hypothesis that asset liquidation values influence both firm leverage and the choice of debt maturity. Using panel data on REITs, we estimate a simultaneous equation model and find that firms specializing in the most (least) liquid assets use more (less) leverage and longer (shorter) maturities. The evidence also suggests that, for REITs, debt maturity and leverage are substitutes, consistent with the theory and predictions of Barclay, Marx and Smith.

Keywords: Asset Liquidation Value Hypothesis, Capital Structure, Debt Maturity, Leverage, Lease Structure, REITs.

JEL Classification: G32

Recent corporate finance literature has emphasized that a firm's financing decisions and capital structure reflect the simultaneous choice of leverage and debt maturity. Firms choose to include debt in their capital structure for a variety of reasons including low all-in cost, tax benefits and the reduction of agency problems associated with free cash flow. However, the use of risky debt creates distortions in the real operations of firms including the classic under-investment problem which predicts that managers of firms with risky debt may pass up positive net present value investment opportunities. Theory (e.g., Myers, 1977 and Hart, 1993) shows that such distortions can be mitigated through the choice of maturity and that consequently the choice of leverage and maturity must be made jointly considering both the costs and benefits of each component of the capital structure. However, any use of risky debt raises the possibility of financial distress and suboptimal liquidation of the firm's assets. These costs, in combination with market forces in the securities markets, can limit management's flexibility to choose certain combinations of leverage and debt maturity, thereby limiting its ability to use debt to influence agency problems. Shleifer and Vishny (1992) and Benmelech (2005) show that, in equilibrium, firms with less liquid assets will use lower leverage and shorter maturity because of the costs associated with suboptimal liquidation. This paper provides empirical evidence supporting the Shleifer and Vishny (1992) hypothesis using a sample of real estate investment trusts (REITs) for which the variation in liquidation value can be measured.

While the importance of agency issues in capital structure decisions has been documented in several recent empirical studies (e.g., Barclay, Marx and Smith, 2003, Johnson, 2003 and Datta, Iskandar-Datta and Raman, 2005), evidence on the importance of asset liquidation value has been limited by the difficulty in quantifying liquidation value. Johnson (2003) estimated a system of equations for both leverage and debt maturity, focusing primarily on the relationship between capital structure and the under-investment problem. He finds that, as predicted by Myers (1977) and Hart (1993), firms with

substantial growth opportunities use lower leverage and shorter maturity debt. In this paper, we extend Johnson's (2003) model to include the effect of the liquidation value of a firm's assets on its choice of capital structure, controlling for growth opportunities and other factors. We find support for Shleifer and Vishny's (1992) and Benmelech's (2005) predictions that firms with less liquid assets will choose to use lower leverage and shorter maturity.

The remainder of this paper is organized as follows. The next section discusses the underlying theory establishing a relationship between leverage and debt maturity resulting from a firm's efforts to mitigate the under-investment problem and reduce deadweight bankruptcy costs. The following section discusses the measures of liquidation value used in the models. The fourth section describes the models to be estimated, the variables used and the data. The last two sections of the paper present the results and the conclusion.

Capital Structure Theory

Since the seminal work of Modigliani and Miller (1958), corporate finance literature has developed a number of different theories to explain the observed cross-sectional variation in capital structure. Fama and French (2002) focus on two major paradigms: trade-off theory and pecking order theory. The trade-off theory argues that firms balance the benefits of debt (primarily low all-in cost, tax shields and the reduction of agency costs associated with free cash flow) against the costs of debt (including under-investment, bankruptcy costs and agency costs arising from conflicts between stakeholders). An alternative (and, in part, complementary) explanation is provided by Myers' (1984) pecking order theory which attributes variation in capital structure to different levels of asymmetric information associated with different sources of capital. Both theories consider capital structure as

multidimensional (including both leverage and maturity) with principal-agent problems, asymmetric information problems and conflicts between stakeholders playing prominent roles.

From the outset, the underlying agency theory indicated a significant relationship between the leverage and debt maturity dimensions of capital structure and growth opportunities. For example, Myers (1977) and Hart (1993) provide much of the underlying theory for the prominence of agency issues. When a firm's investment policies are not fixed, different capital structures alter the incentives for shareholders and managers, sometimes inducing managers to forego positive net present value projects (under-invest) and sometimes to undertake risky negative net present value projects (over-invest). Both problems are best understood by considering that the equity in a firm with risky debt can be viewed as a call option on the firm's assets with a strike price equal to the face value of the debt. Since the value of a call option is attributable solely to the payoffs above the strike price, in the presence of risky debt, the shareholders receive only a portion of the market value of the payoffs from a new investment project funded with internally generated funds. Consequently, such shareholders will rationally forego certain positive net present value projects when the cost exceeds their share of the benefits. Furthermore, the higher the strike price (i.e., face value of the debt) the more significant is the under-investment problem and thus the under-investment problem can be reduced by using less debt. However, as modeled by Hart (1993), the under-investment problem can also be reduced by using shorter maturity debt because if the debt matures before the payoff from the investment is realized, the problem of sharing the payoff is eliminated. Thus, the theory put forth by Myers (1977) and Hart (1993) predicts that firms with significant investment (growth) opportunities will use lower leverage and shorter maturity debt than firms with few growth opportunities.

Recent corporate finance literature has begun to address the importance of the multidimensionality of a firm's capital structure by studying the simultaneous choices of leverage and debt maturity. Barclay,

Marx and Smith (2003) show that under certain circumstances, longer maturity debt can be substituted for higher leverage and vice versa because both long maturity debt and higher leverage work to limit management's flexibility to use free cash flow to over-invest. Since both higher leverage and longer maturity also contribute to the under-investment problem, the optimal capital structure must balance these costs and benefits and will likely involve the balancing of maturity and leverage.

Inherently, the use of risky debt entails some risk of default and liquidation in certain states of the world. Therefore, market forces and liquidation costs will affect a firm's ability to freely exploit the benefits of leverage and debt maturity. Williamson (1988) and Shleifer and Vishny (1992) provide a theory linking the liquidation value of a firm's assets with the firm's choice of capital structure. Williamson (1988) made the point that assets which are easily redeployed into alternative uses will have high liquidation values and are therefore easier to finance. Lenders know that if a borrower is unable to make the contractual debt payments and the lender has to liquidate the firm's assets, they are better able to recover their investment if the assets retain their value in their next best use. Shleifer and Vishny (1992) extended this idea by considering the general equilibrium impact of asset liquidation value on firm capital structure. The highest liquidation price will normally be attained when a firm's assets are sold within its own industry.¹ However, it is frequently the case that when financial distress strikes one industry participant, other participants will also experience financial distress and be unable to raise the capital needed to bid for assets being liquidated and as a result assets that are not easily redeployable will be sold at low prices. Because industries differ in the degree to which major participants' financial fortunes covary and the degree to which their assets can be redeployed to a new use, there will be significant variation in equilibrium capital market structures across industries.

¹ Brown (2000) provides evidence consistent with this prediction comparing market performance for equity and mortgage REITs during the real estate market downturn in the late 1980s.

Shleifer and Vishny (1992) predict that firm debt capacity (leverage) in an industry should fall as liquidation value falls. They also show that, in equilibrium, industry debt capacity reflects a balance between the benefits of debt in reducing principal-agent problems and the costs associated with losses from the sale of assets to firms outside the industry in the event of financial distress. For example, firms in cyclical industries are likely to share in prosperity and recession and, as a result, industry assets will be less liquid because it is likely that a liquidating firm's assets will have to be sold to industry outsiders at lower values.² Although Shleifer and Vishny (1992) do not directly model debt maturity, Benmelech (2005) points out that their equilibrium model depends on the creation of a long-term debt overhang so that a firm cannot borrow against future earnings to finance negative net present value projects. Thus, the Shleifer and Vishny (1992) model also predicts that debt maturity will be positively correlated with liquidation value.

The literature testing their theoretical predictions has been limited, especially in the context of a firm's capital structure. Two previous empirical studies have provided evidence that liquidation value is significant in financing decisions. Benmelech (2005) studies 19th century railroad debt policies and finds that railroads with rolling stock that was more readily redeployable by other firms in the industry used longer maturity debt, but found no evidence of higher leverage for those firms. Benmelech, Garmaise and Moskowitz (2005) study commercial mortgage loans and find that properties with greater zoning flexibility were financed with loans that had longer maturity, higher loan-to-value ratios and lower interest rates.³

² Brown, Ciochetti and Riddiough (2006) study the resolution of financial distress for an owner-managed project such as a real estate asset. They show that asset liquidation value is a significant factor in the lender's decision to foreclose or restructure a loan and consequently also influences the borrower's decision to default. Their model provides an alternative motivation for selling assets to new managers than does Shleifer and Vishny (1992), namely by selling the asset to more productive outsiders, the lender shares in some of the gains from improved investment incentives.

³ Benmelech, Garmaise and Moskowitz (2005) develop an index of redeployability based on detailed, manually collected information on the zoning regulation for each property location. Their index takes into consideration both the zoning category (e.g., industrial, apartment etc.) and a numeric measure of "within category" intensity where higher numeric values reflect fewer restrictions on the potential property development.

In this paper, we empirically test two predictions from Shleifer and Vishny (1992) and Benmelech (2005). The first prediction is that firms with assets that have lower liquidation values will use lower leverage. The second is that debt maturity is positively related to asset liquidation value. In addition, by estimating a simultaneous equation system that includes liquidation value, we provide new information on whether maturity and leverage are complements or substitutes. The REIT industry provides an ideal opportunity to test the Shleifer and Vishny (1992) theory because REITs tend to concentrate on a single property type and the liquidity and redeployability of commercial real estate assets vary systematically by property type—much like the industry-to-industry variation discussed by Shleifer and Vishny. The five major property types (industrial properties, residential apartments, hotels, office buildings and retail shopping structures) represent five distinct lines of business with unique economic sensitivities. REITs are regulated entities that are essentially restricted to owning real-estate assets. If a REIT follows the specified investment and earnings distribution rules established by Congress, its earnings are effectively exempt from taxation at the corporate level.⁴ The homogeneity of REIT regulations and operations reduces the need for control variables when modeling capital structure. For example, there is no need to control for net operating loss carry forwards or other tax related factors which have been included in most cross-sectional studies of capital structure.

⁴ In order to maintain its tax-advantaged status as a REIT, the firm must adhere to regulations limiting its investments to real estate and certain other assets and distribute 90 percent of all pretax earnings (95 percent prior to the 1999 REIT Modernization Act) as dividends. For a full discussion of the regulations REITs must follow, see Chan, Erickson and Wang (2003) and Brueggeman and Fisher (2005).

Measuring Liquidation Value

To test the Shleifer and Vishny (1992) predictions, we extend the Johnson (2003) model by including a proxy for the liquidation value of the firm's assets. The following subsections describe three different approaches to measuring liquidation value.⁵

Lease Maturity. Brown and Riddiough (2003) suggest a linkage between the average debt maturity of a REIT and its lease structure. A REIT earns income by leasing the space in the properties it owns. The lease maturity structure is of critical importance to the future income and operating strategy of a REIT. Geltner and Miller (2001) describe the bargaining process between lessor and lessee that determines the equilibrium lease maturity for a particular property type and market. That equilibrium lease maturity reflects a balance among a variety of factors including the tenant's cost of relocating, the landlord's cost of releasing (including expected vacancy time) and the outlook for supply and demand in the property-type sector. In turn, the resulting lease maturity structure influences the operating strategy and financing policy for the REIT.

Because a REIT that enters into long-term leases for its properties has, in theory, "locked in" future cash flows from its tenants, one might think that the REIT managers should similarly "lock in" its financing costs by using longer maturity debt. However, most long-term commercial lease contracts provide for some form of inflation adjustment protection and protection against escalating operating costs that reduce the importance of "matching" maturities. For properties where the market equilibrium dictates very short maturity leases (e.g., hotels and apartments), the high cost and risk of

⁵ In addition to the three measures discussed here, we tested the average property size in the REIT's portfolio as a measure of liquidation value. Traditionally, unusually large properties have been viewed by industry participants as being less liquid. The analysis was complicated by missing data for several REITs and the need to exclude small numbers of "outlier" REITs (e.g., retail REITs specializing in regional malls). Furthermore, because our unit of observation was a REIT, we had to use the average size of all properties in the REIT portfolio. This averaging removed a lot of the variability and made it difficult to estimate the effect precisely. As a result, we have not included those results in the paper. Nevertheless, we believe property size is a potentially significant variable to develop for future research on liquidation value.

rolling over short-term debt financing reduces the benefits of maturity matching. Furthermore, the factors that drive changes in rent equilibrium are significantly different than the factors that influence market interest rates and so matching lease maturity with debt maturity is quite different than matching the maturity of financial assets with the maturity of debt. However, the lease structure of a REIT also influences the operating flexibility or option value associated with the ownership of the real estate and that option value is related to the liquidation value of the property.

Short-term leases provide an option to release at more favorable conditions for both landlord and tenant regardless of their expectations about future short-term rents. By signing long-term leases, both parties give up the value of this option. Long-term leases also restrict an owner's ability to modify the use of a property. For example, commercial real estate is often described by a "grade" with A grade properties being the highest quality and attracting the best tenants and highest rents, but requiring higher levels of maintenance and improvements. Depending on local competition and demand for space, an owner may choose to upgrade a property from B to A by renovation and improvements or downgrade the property through lower expenditures on maintenance and improvements. Long-term leases restrict an owner's flexibility to improve the grade of the property because rents would continue to reflect the original B quality for a long time. Similarly, long-term leases restrict an owner's ability to redeploy the asset to an alternative use (e.g., from light industrial to retail). When a property experiences financial distress, its liquidation value may be determined by bidders considering an alternative use for the property. As a result, we expect that properties encumbered with longer term leases will have lower liquidation values attributable to the limits on redeployment associated with those longer-term leases.⁶

⁶ This prediction implicitly assumes that the leases entered into by the property owner remain in place when the asset is liquidated or alternatively, that average lease maturity today is a good predictor of average lease maturity in financial distress. If the financial distress of the REIT was caused by tenants breaking their leases and leaving the property, the average lease maturity today may not be a good predictor of average future lease maturity. However, tenants cannot break

An alternative view of lease maturity leads to the same relationship between lease maturity and liquidation value.⁷ Anecdotal evidence suggests that when properties are built for a specific tenant with unique needs, the owner will demand a longer lease commitment. This observation suggests that long maturity leases are the equilibrium outcome for assets that are difficult to redeploy to alternate uses. Thus, although the cause and effect relationship are reversed here, the conclusion is the same – properties with longer maturity leases are likely to have lower liquidation values.

REITs exhibit significant variation in average lease maturity by property type focus, ranging from a low of one day for hotels to approximately seven years for retail properties. In our sample, industrial and office properties have average lease maturities of four and six years, respectively, while apartment REITs have lease maturities that average roughly one-half year.

Recovery in Foreclosure. The recent rapid growth of Commercial Mortgage-Backed Securities (CMBS) provides an opportunity to observe actual liquidation values of properties experiencing financial distress. A CMBS issue is generally collateralized by a diversified pool of commercial mortgage loans. When a loan becomes seriously delinquent, the servicing of the loan is transferred to a special servicer charged with maximizing the recovery of value. When the servicer concludes that the current borrower is unlikely to be able to repay the loan, the underlying commercial property is foreclosed and sold. Standard and Poor's (2005) provides a detailed analysis of the recovery rates on such loans by property type over the period from 2000 through 2004. The report shows recovery rates (as a percent of original loan amount) of 70.2% for residential apartment loans, 61.1% for office properties, 60.5% for industrial properties, 59.8% for retail loans and 51.5% for loans secured by hotels and motels. While these specific recovery rates are sensitive to the economic conditions during

leases with impunity and past history has shown that financial distress of REITs is more attributable to market forces such as declining rents, rising operating expenses or declining property values.

⁷ We are grateful to an anonymous referee for suggesting this line of reasoning.

the observation period, the average recovery rates and ranking with apartments showing the highest recovery rates and hotels the lowest with other property types falling in the intermediate range is consistent with data from the 1980s based on life insurance company experience (see Snyderman, 1991, 1994). Although it is common to classify REITs by their property focus as industrial REITs or office REITs, they are permitted to invest in more than a single property type. As a result, we calculate a weighted average portfolio recovery rate using the S&P property type recovery rates and the property type mix reported for the REIT's portfolio. Because most REITs have the vast majority of their assets in the primary property type, there is relatively little variation in this measure from REIT to REIT within a property type. The liquidation value hypothesis predicts that REITs that specialize in properties with higher liquidation values will use higher leverage and longer debt maturity.

Multidimensional Ranking. Because asset liquidity is a multidimensional attribute, we also consider a third approach to measuring liquidation value based on four different factors: zoning, physical flexibility and (the previously discussed) lease maturity and recovery rates.

Zoning regulations can establish a pecking order of uses ranging from residential (considered the highest use with fewest negative externalities) to industrial (lowest use with the most negative externalities). For a more complete discussion of the underlying theory and economic effects of zoning, see Fischel (1985). In many jurisdictions, land can be shifted to an alternative use ranking higher in the pecking order. For example, land zoned for industrial use can often be redeveloped into office, retail or residential use, but land zoned residential generally cannot be shifted to commercial use without time-consuming and costly rezoning efforts. We use this ordering of zoning flexibility as a third dimension of liquidity. Based on the zoning pecking order, residential apartments have the least

zoning flexibility and industrial properties have the greatest zoning flexibility. The other property types fall in between these two and are ranked as intermediate.⁸

The structural attributes of various types of real estate also affect the liquidity of real estate assets. For example, an industrial property is often simply a big open structure used as a warehouse or distribution center. If one tenant leaves, it is generally physically easy to reconfigure the space for a different user from a different industry. Reconfiguring industrial space for a new tenant is relatively inexpensive. Furthermore, the simplicity and flexibility of the basic industrial property structure reduces the cost of converting the space to an alternative use such as low-rise suburban office space. (For more discussion about this topic, see Society of Industrial Realtors, 1984.) On the other hand, high-rise office space is generally quite expensive to reconfigure. The finish level and physical floor configuration make it costly to even change tenants. For example, the Urban Land Institute (1982) describes in its publication, *Office Development Handbook* that, in office design, the location of the columns and building core (elevators, stairs, mechanical and plumbing) must be carefully planned to achieve the desired office layout. The resulting specialized physical structure and location in high density areas constrain the owner's ability to redeploy this type of asset.

Similarly, retail shopping structures, hotels and apartment buildings have their own unique design features tailoring them for the intended use. For example, retail stores depend on visible "frontage" to draw customers into the store. Therefore, retail structures are designed with wide spacing between structural columns and shapes that maximize frontage. These design features limit the redeployability both within the industry and to alternate uses. (For more discussion of the unique design features of

⁸ The Benmelech, Garmaise and Moskowitz (2005) index of zoning flexibility provides a finer measure of redeployability and is in theory preferable to the coarse indicator used here. However, we do not have the micro level property location and zoning information detail needed to create a similar index. Our measure of zoning flexibility focuses on the ability of an owner to shift a property to a use outside its current "industry" or property type usage roughly corresponding to the first dimension used by Benmelech, Garmaise and Moskowitz (2005).

retail properties, see Urban Land Institute, 1977 and White and Gray, 1996.) Additional design and development information for apartment buildings and hotels is provided in Urban Land Institute (2000) and Geltner and Miller (2001). Based on these industry design and development principles, office buildings provide the least flexible physical structures while industrial properties provide the greatest flexibility. The remaining property types fall between these two on the scale of physical flexibility.

Table 1 summarizes the discussion of all four dimensions of liquidity. The table has three columns to reflect three different levels of liquidity: High, Intermediate and Low. Based on the evaluations discussed above, we assign the property type to its appropriate cell. For the two quantitative dimensions, the assignment is based on the quantitative ranking, using natural “breaks” in the measure to identify the three categories. For example, both residential apartments and hotel properties are assigned to the High category for lease maturity because both property types have relatively short-term lease contracts (less than one year). The category assignments for the qualitative measures (zoning and physical flexibility) are based on the discussion above. In each case, one property type is assigned to the High category, one to the Low category and the remaining three were assigned to the Intermediate category.

Table 1 shows that retail and office properties rank Low on one dimension (physical flexibility for office and lease maturity for retail) and Intermediate on the other three dimensions. Industrial properties rank High on two dimensions (zoning and physical flexibility) and Intermediate on two (recovery rate and lease maturity). Apartment properties, like industrial properties, rank High on two dimensions, but rank low on zoning. Overall, the table indicates that, based on this multidimensional ranking, industrial REITs have the highest liquidity ranking, followed closely by apartment REITs. Retail and office REITs have the lowest overall ranking on this measure while hotel REITs fall in the intermediate category. When estimating the capital structure model with this measure of liquidation

value, we use indicator variables for property type to control for liquidation value and expect a significant difference in coefficients on the indicator variables for industrial (high liquidity) and retail and office properties (low liquidity).

Model Specification and Data

We follow standard practice (as in Johnson, 2003, and Stohs and Mauer, 1996) in selecting proxies for other traditional factors used in empirical studies of capital structure. These factors, and their hypothesized effects, are summarized in Table 2. As suggested by Myers (1977) we use the market-to-book ratio to measure growth opportunities, where the market value of assets equals the book value of assets plus the difference between market value of equity and book value of equity (measured in billions of 1997 dollars⁹). Diamond (1991) argues for a non-linear relationship between firm quality and debt maturity. We use firm size (measured in billions of 1997 dollars) and the volatility of cash flow (measured by the time series standard deviation of funds from operations) as proxies for firm quality.¹⁰ Following Johnson (2003) and Stohs and Mauer (1996), we use the forward-looking actual growth rate of earnings $((EPS_{t+1}-EPS_t)/Share\ Price_t)$ to proxy for the asymmetric information effect that creates a tendency for firms with improving prospects to use shorter maturity debt. To control for Myers' (1984) pecking order prediction that profitable firms will use less debt, we include a measure of profitability. We measure profitability using the return on assets and use an indicator variable that takes on a value of one if the firm has a public credit rating as a proxy for market access. We adopt the convention of Stohs and Mauer (1996) and Johnson (2003) and measure asset maturity (in years) as the ratio of the dollar value of real estate assets to depreciation expense.

⁹ We use the Producer Price Index (PPI) published by the U.S. Department of Labor as the deflator.

¹⁰ Funds from operations is defined as net operating income from the properties owned by the REIT. Net operating income is essentially the difference between rental income and building operating expenses. It measures net cash flow from the properties and does not include interest expense, depreciation or corporate taxes.

Our base “traditional” simultaneous equation system has the form:

$$\text{Leverage} = \beta_0 + \beta_1 \text{Ln}(\text{Debt Maturity}) + \beta_2 \text{Market - to - Book} + \beta_3 \text{Ln}(\text{Firm Size}) + \beta_4 \text{Volatility} + \beta_5 \text{Return on Assets} + \beta_6 \text{Market Access} + \varepsilon_L \quad (1)$$

$$\begin{aligned} \text{Ln}(\text{Debt Maturity}) = & \gamma_0 + \gamma_1 \text{Leverage} + \gamma_2 \text{Market - to - Book} + \gamma_3 \text{Ln}(\text{Firm Size}) + \\ & + \gamma_4 \text{Ln}(\text{Firm Size})^2 + \gamma_5 \text{Volatility} + \gamma_6 \text{Earnings Growth} + \gamma_7 \text{Asset Maturity} + \\ & + \gamma_8 \text{Market Access} + \varepsilon_{DM} \end{aligned} \quad (2)$$

In the above equations, leverage is defined as the ratio of total book value of all debt to the market value of the firm’s assets and debt maturity is measured in months using information reported by SNL Data Source.¹¹ As in Johnson (2003), the system is identified by the exclusion of firm size squared, earnings growth and asset maturity from the leverage equation and the exclusion of return on assets from the maturity equation. Firm size squared is excluded from the leverage equation because the relationship between leverage and firm quality is expected to be linear (see Diamond, 1991). Earnings growth and asset maturity are excluded because these variables are expected to influence only the maturity choice and not the leverage. Profitability is excluded from the maturity equation because current profitability is not expected to influence maturity after controlling for earnings growth.¹²

After estimating the base model, we re-estimate the system by sequentially adding the three different measures of liquidation value into both equations. For the specification with lease maturity, we

¹¹ The reports provide the dollar amount of debt maturing in each of the next five years and the total of debt maturing in six years or longer. We calculate a weighted average maturity using the percentage of the debt maturing in each category and the upper limit of the category as the maturity for all debt in that category. For example, if a firm with \$40 million of total debt had \$10 million of debt maturing in the next 12 months and \$30 million of debt maturing in months 37-48, we would calculate the average maturity as $(10 \times 12 + 30 \times 48) / 40 = 39$ months. We assign a maturity of six years to all debt in the six years and longer category. Previous papers have used simplified measures of debt maturity such as the percentage of debt maturing after a specified time. As a robustness check, we have estimated our model using the percentage of debt maturing in five years or more as the dependent variable in equation (2). Our results are qualitatively similar.

¹² In order to study the impact of growth opportunities in greater detail, Johnson (2003) includes the interaction of debt maturity and market-to-book in his leverage equation. The inclusion of the interaction term enables Johnson to test whether the use of short term debt reduces the negative effect of growth opportunities on leverage. He finds this to be true. We are unable to use this approach with our single industry data because the variation in market-to-book ratio dominates the variation in the predicted debt maturity from our first stage regression. The resulting interaction term is too highly correlated with market-to-book.

collected information on average lease maturity from a variety of sources. For retail, office and industrial REITs, the data was obtained from SNL Data Source. For apartment REITs, the data was collected from the firm's 10-K. For hotel REITs, the average lease maturity was set to one day. For the specification with recovery value, we used the average recovery data reported by S&P for each property type and calculated a weighted average recovery value for each observation in the sample using the percentage of the REIT's overall portfolio allocated to each property type in the given year as the weights.¹³ For the multidimensional measure of liquidation value, we include property type indicators. We use apartment REITs as the excluded group so the coefficients should be interpreted as indicating a shift relative to apartment REITs.

The initial sample consists of the 136 publicly traded equity REITs reported in the SNL Data Source database as of December 31, 2003 that specialize in one of the five major property types.¹⁴ The balance sheet and income statement items required for the estimation were obtained from COMPUSTAT and other variables from SNL Data Source. The sample period ranges from 1997 to 2003. We chose 1997 as the starting year for data collection because data for some key variables used in the analysis are not available in COMPUSTAT or SNL Data Source prior to 1997. We terminate the panel with 2003 because, at the time of the analysis, the earnings growth control variable was only defined through 2003. The sample size is reduced because of missing data on average debt maturity or average lease maturity, so the final sample includes 57 equity REITs for which we have one or more complete observations over the sample period and 193 pooled time series, cross-sectional observations.

Table 3 provides summary statistics for the variables used in the estimation. The table shows that office REITs represent the largest portion of the sample, closely followed by retail REITs. In

¹³ The results using the weighted average recovery rate do not differ materially from the results obtained if the recovery rate were determined by the REIT's primary property focus because most REITs have close to 100% of their real estate assets invested in their primary property type.

¹⁴ This excludes REITs that specialize in health care, manufactured housing, self-storage facilities and diversified REITs.

combination, these two types represent seventy-two percent of the sample. Apartment REITs represent approximately fifteen percent. Hotel REITs represent less than five percent of the sample.

The REITs in our sample use more leverage and longer maturity debt than do the primarily industrial firms studied by Johnson (2003), Stohs and Mauer (1996) or Datta, Iskandar-Datta and Raman (2005). All REIT types, except hotel REITs, have average market-to-book ratios in excess of one and positive average return on assets. Roughly half the sample has public credit ratings.

Results

Table 4 presents the results of estimating the base model (equations (1) and (2)) that contains only the traditional corporate finance explanatory factors using two stage least squares.¹⁵ The table shows that REITs with significant growth opportunities, as proxied by market-to-book ratio, use lower leverage and shorter maturity debt. This finding is consistent with Myers' (1977) and Hart's (1993) predictions that firms with significant growth opportunities will use lower leverage and shorter maturity debt to avoid the under-investment problem and with both Barclay, Marx and Smith's (2003) and Johnson's (2003) empirical results. Also similar to the findings reported by both empirical studies, firm size enters the leverage equation with a significant negative effect. Firm size and firm size squared enter the leverage equation, but the significance of the estimated effects is marginal. Consistent with Myers' (1984) pecking order theory, highly profitable firms use lower leverage. The only unusual result is the positive relation between volatility and leverage and Hart (1993) provides a rationale for volatility to potentially have such an effect.

Turning next to the endogeneity of debt maturity and leverage, we see that both coefficients are negative. Long maturity debt is associated with lower leverage. The coefficient on debt maturity in

¹⁵ The first stage regressions as well as estimations discussed in the robustness section are available upon request from the authors.

the leverage equation is significant at the five percent level. The coefficient on leverage in the maturity equation is also negative but is not significant. This combination of coefficients suggests that short debt maturity and low leverage are substitutes. This result is consistent with the prediction of Barclay, Marx and Smith (2003). These authors show that the coefficient on leverage in the maturity equation and that on maturity in the leverage equation should have the same sign. They further show that the sign depends on whether maturity and leverage are substitutes (negative signs) or complements (positive signs). Barclay, Marx and Smith argue that because leverage and maturity can each be used to control or reduce the underinvestment problem, they should be viewed as substitutes. Johnson (2003), however, argues (based on Diamond, 1991, 1993 and Sharpe, 1991) the opposite point of view – that long term debt and high leverage are complements. Johnson (2003) argues that firms that use high leverage are taking on greater risk of bankruptcy and costly liquidation and that such firms tend to avoid the additional liquidity risk associated with rolling over short term debt by using long-maturity debt. Johnson (2003) finds positive coefficients¹⁶, consistent with his argument for complementarity while Barclay, Marx and Smith find opposite signs on the two endogenous variables.¹⁷ For our sample of REITs, the coefficients support the substitution theory proposed by Barclay, Marx and Smith (2003). Whether maturity and leverage are substitutes or complements could well vary from industry to industry or even firm to firm based on whether the debt was being used primarily to limit the underinvestment risk as discussed by Barclay, Marx and Smith or whether the firm's primary concern was the layering of risks discussed by Johnson. As noted earlier, REITs tend to have high levels of tangible assets because they are restricted to owning real estate and Treasury securities. They also

¹⁶ Johnson uses the percentage of total debt that matures in the next three years as his measure of maturity which means that as average debt maturity increases, his "maturity" measure decreases. In our model, as in most previous studies of debt maturity, the dependent variable measuring maturity increases with greater use of long maturity debt. The discussion in the text reflects this more conventional definition of maturity and consequently we have reversed the signs actually reported by Johnson for consistency.

¹⁷ Barclay, Marx and Smith (2003) describe this result as "puzzling" and suggest that it might be attributable to model misspecification such as the exclusion of other capital structure policy variables such as leasing, hedging and dividend payout policy.

operate with much longer maturity debt and higher leverage than the typical industrial firm. Given the well-documented corporate governance issues associated with the REIT structure (see Ghosh and Sirmans, 2006) and the very liquid mortgage market available to them for financing, it may well be that using debt as an internal governance mechanism is a more significant concern for REITs than the rollover risk described by Johnson (2003). In turn, this could explain the apparent dominance of the substitution effect in REIT capital structures.

Table 5 presents the results for the extended model that includes lease maturity as the proxy for liquidation value. As expected, lease maturity enters both the leverage and debt maturity equations with significant negative coefficients. This is consistent with the arguments that long maturity leases are associated with properties that have lower liquidation values. The point estimates of the coefficients on the traditional control variables are similar in sign and magnitude to those reported in Table 4, however they are estimated somewhat less precisely and only the market-to-book coefficients are statistically significant. This suggests that there is some relationship between the traditional proxy variables and lease maturity and reinforces the importance of controlling for liquidation value in models of capital structure.¹⁸ The signs on the two endogenous variables are still both negative and the coefficient on leverage in the maturity equation is now weakly significant.

Table 6 reports the results using the weighted average recovery rate as the proxy for liquidation value rather than average lease maturity. Table 6 shows that the estimated coefficients on the recovery index both have the expected positive sign, but only the coefficient in the maturity equation is statistically significant. As with Table 5, the point estimates of the coefficients on the traditional control variables

¹⁸ Lease maturity is not highly correlated with any of the individual traditional variables. The highest correlation coefficient is with the volatility of funds from operation (-.37).

are similar to those reported in Table 4, but the significance levels in Table 6 are closer to those originally reported in table 4.¹⁹

Finally, Table 7 presents the results using four property type indicators as the proxy for liquidation value.²⁰ The coefficients on the four property type indicators are consistent with Shleifer and Vishny's (1992) liquidation value theory. With apartment REITs being the excluded property type, a significant coefficient indicates a significant difference between the effect of that property type and apartment REITs. As expected, office and retail REITs, which were identified in Table 2 as having the lowest liquidation values, both have negative coefficients in both equations. The office coefficient is negative and significant at the one percent level in both equations, indicating that office REITs use lower leverage and shorter debt maturity than apartment REITs, after controlling for the traditional capital structure effects. The retail coefficient is weakly significant in the maturity equation only. Because industrial REITs were identified in Table 2 as having the most liquid assets (based on the multidimensional index) while office REITs had the least liquid assets, it is of interest to test the coefficients on these two indicators for equality. The coefficient on the industrial indicator is significantly different (larger) from the office coefficient at the one percent level in the maturity equation and the five percent level in the leverage equation. The significant differences in these coefficients are consistent with the prediction that REITs with less liquid assets would use lower leverage and shorter maturity debt. Industrial REITs, *ceteris paribus*, use higher leverage and longer debt maturity than do office REITs. The coefficients on the endogenous variables are similar to those reported in Table 4 and are still consistent with debt maturity and leverage being substitutes. The

¹⁹ The correlations between the recovery index and the traditional variables are somewhat lower but generally similar in magnitude to those of the lease maturity. For example, the correlation between the recovery index and the volatility of funds from operations is -.28.

²⁰ Given the definition of lease maturity and weighted average recovery rate, it is impossible to include both the property type indicators and either of the other proxies for liquidation value. The variables would be too highly correlated to obtain separate estimates of the individual effects. In the model specification reported in Table 7, it is impossible to be certain whether the estimated effect is the result of differences in liquidation value or other differences across REIT type.

estimated coefficients on the traditional control variables are essentially the same as those discussed in Table 4, but the standard errors for the effects other than growth opportunities are larger, reducing the t-statistics.

To gain an overall perspective on the results, we review the main results from all four specifications. All three specifications that include a proxy for liquidation value show a significant effect of liquidation value with a sign consistent with the Shleifer and Vishny (1992) theoretical predictions. The significance of this consistency is reinforced by the fact that the three different proxies for liquidation value are not highly correlated and potentially measure different aspects of liquidation value. For example, the office indicator has correlation coefficients of .31 and -.10 with lease maturity and the recovery index, respectively. A logit model for the office indicator based on the other two liquidation value proxies has a pseudo R^2 of approximately ten percent.

The combination of models also shows that controlling for liquidation value reduces the significance of the other variables traditionally included in models of capital structure. Most of these traditional control variables are significant (at least marginally) in the model without a control for liquidation value. Only the market-to-book ratio is significant in all three specifications with controls for liquidation value. Furthermore, while debt maturity is robustly significant in all four models as a determinant of leverage, leverage is only marginally significant in the model with lease maturity as a proxy for liquidation value. These findings are worthy of further exploration in a larger sample of firms from different industries if a suitable measure of asset liquidation value can be developed.

Robustness

The fact that all three specifications using different proxies for liquidation value generate essentially the same result provides strong evidence that the liquidation value of the underlying assets influences

REIT capital structure.²¹ Liquidation value and growth opportunities appear to have the greatest and most robust influence on REIT capital structure. Our results further suggest that controlling for liquidation value is important in general models of capital structure.

As a first robustness check, we re-estimated all four models using three-stage least squares, allowing for correlation of the residuals from the two separate equations. The results are essentially the same as those reported in Tables 4-7. The coefficients on the liquidation proxies retain the correct signs and have similar statistical significance. In fact, the estimated p-values for the liquidation proxies are lower in several of the three stage least squares runs. For example, in the lease maturity specification, the coefficient on lease maturity is significant in the leverage equation at the five percent level (versus the ten percent level in the two stage least squares model) and, in the property-type specification, the retail indicator is significant at the one percent level in the maturity equation (compared with the ten percent level in the two stage least squares model). Market-to-book ratio remains strongly statistically significant in all specifications and enters with the expected negative sign. The coefficients on the endogenous variables, leverage and debt maturity, are both negative (consistent with the substitution argument) and have essentially the same magnitudes and statistical significance as reported using two stage least squares. In certain specifications, some of the traditional variables that were marginally significant in the two stage least squares estimates lose their significance when estimated using three stage least squares.²²

²¹ The coefficients are economically significant as well as statistically significant. For example, based on the model using lease maturity as a proxy for liquidation value (Table 5), a one standard deviation increase in lease maturity reduces the predicted leverage ratio by .03 (30% of the standard deviation of leverage) and reduces the predicted debt maturity by 6.5 months (19% of the standard deviation of debt maturity). The economic significance results are even stronger if we use the model that uses property type indicators (Table 7) and slightly weaker if we use the model based on recovery rate (Table 6).

²² We do not report the three stage least squares results as our primary results because our sample size is relatively small to depend on the asymptotic properties of three stage least squares.

Next we re-estimated the models using an alternative definition of debt maturity. For these estimations, we defined debt maturity as the percentage of all debt maturing in five years or more.²³ Our findings with respect to the importance of liquidation value and the signs of the coefficients of the endogenous variables are the same with the alternative definition of debt maturity. Finally, we estimated a model that included both lease maturity and recovery index.²⁴ The estimated coefficients on both liquidation value proxies are identical in sign and quite similar in magnitude. However, as econometric theory predicts when one includes correlated variables, the precision is lower in the combined model.

Because the same firm can enter our pooled sample data several times, it is likely that the standard errors are biased downward in Tables 4-7.²⁵ To test for robustness of our major findings to this possibility, we re-estimated the models averaging out the time series aspect by using firm-specific time series means for all variables and estimating a purely cross-sectional model. This averaging process significantly reduces the variability in the data and reduces the sample size to fifty-seven observations. Both of these results make it more difficult to estimate effects precisely. In the specification with only the traditional variables (as in Table 4), none of the variables are statistically significant (in either equations). However, the average lease maturity is significant at the five percent level and the recovery index is significant at the ten percent level in the maturity equation. In the property type specification, the office indicator enters the leverage equation with a negative coefficient that is significant at the five percent level. Overall, these results indicate that the liquidation value effect is quite strong and robust. We also estimated the standard errors of all four models using STATA's

²³ While several other finance studies have used the percentage of debt maturing three years or more in the future, REITs have much longer average maturity of debt and the longer cutoff is more appropriate for this sample.

²⁴ It is inappropriate to estimate a single model that includes all three proxies in the same model because, by construction, the lease maturity and recovery index are almost perfectly correlated with the property type indicators.

²⁵ Because the liquidation value proxies are constant across all time periods for each firm, we are unable to estimate a model with firm fixed effects.

robust standard errors, clustering by firm. In these regressions, the market-to-book, lease maturity and property-type indicators retain their signs and statistical significance (albeit generally with higher p-values). The p-values for the recovery index exceed the normal threshold for statistical significance.²⁶

Conclusions

Shleifer and Vishny (1992) and Benmelech (2005) convincingly argue that asset liquidation value should influence capital structure. Specifically, they argue that, in equilibrium, firms in an industry will choose capital structures that balance the agency benefits of debt in resolving principal-agent conflicts with the expected cost associated with the impact of debt on liquidation values and bankruptcy costs. Their model predicts a positive relationship between asset liquidity and both leverage and debt maturity.

We use panel data for REITs backed by different types of commercial real estate to provide the first empirical support for this theory in a corporate finance setting. We extend Johnson's (2003) simultaneous equation model for leverage and maturity by including proxies for asset liquidation value and find strong support for the claim that firms with less liquid assets choose lower leverage and shorter debt maturity. Our results for the traditional agency effects on capital structure are consistent with previously reported research. In addition, we find evidence supporting the Barclay, Marx and Smith (2003) substitution argument for leverage and debt maturity.

²⁶ A referee suggested that the time since the firm's last equity offering would be an important variable if one were studying the dynamic behavior of a firm's capital structure. Although this paper is essentially a cross-sectional analysis of capital structures based on the theoretical model of Barclay, Marx and Smith (2003) and does not consider time series changes, we nevertheless collected information on time since the last equity offering for as many observations in our sample as we could. Because we could not obtain complete information, our sample size would be reduced by approximately 25% if we included this variable. Despite this significant loss in sample size, we re-estimated all three models. Although the reduced sample size resulted in a loss of precision in all specifications, the inclusion of the additional variable did not alter the results for any of the debt maturity equations and resulted in only minor changes in the leverage model with the property type indicators (Table 7). We attribute the latter effect to the fact that the reduced sample also altered the mix of property types in the sample. Our principal finding that asset liquidation value influences management's choice of capital structure in a manner consistent with Shleifer and Vishny's theory was robust to this alternative specification and reduced sample size.

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Table 1
Multidimensional Ranking of Property Types by Liquidation Value

This table ranks the liquidation value (from HIGH to LOW) of the five different REIT asset types (Industrial, Apartments, Hotel, Office and Retail) used in this study with respect to *Zoning*, *Physical Flexibility*, *Lease Maturity* and *Recovery Rate*. Ranking of property types with respect to zoning is based on the principle that land can be redeveloped into any higher use where rankings are based on estimated negative externalities. In general, apartment uses are viewed as having the fewest negative externalities and industrial uses the most. See Fischel (1985). Physical flexibility reflects the cost associated with physically redeploying assets into use with a new tenant or in a different category. Rankings are based on the specialized literature related to the design and development of different property types. See for example, the Handbooks for different property types published by the Urban Land Institute and listed in the References. Ranking by lease maturity is based on reported average lease maturity by either SNL Data Source or the firm's 10-K. Ranking by recovery rate is based on the average reported losses on loans backing CMBS as reported by Standard & Poor's (2005).

	HIGH	INTERMEDIATE	LOW
Zoning	Industrial	Retail Office Hotel	Apartment
Physical Flexibility	Industrial	Apartment Retail Hotel	Office
Lease Maturity	Apartment Hotel	Industrial Office	Retail
Recovery Rate	Apartment	Industrial Office Retail	Hotel

Table 2
Traditional Factors Influencing Capital Structure

This table summarizes the variables traditionally used in empirical studies of leverage and debt maturity. A blank in a column indicates that the variable is excluded from that model and shows the restrictions on the coefficients used to identify the system of equations.

	Leverage	Maturity
Traditional Corporate Finance Factors		
<i>Growth Opportunities</i>		
Proxy: Market-to-Book Ratio	-	-
<i>Firm Quality</i>		
Proxy: Firm Size	+	+
Proxy: Firm Size Squared		-
Proxy: Volatility of Earnings	-	+
<i>Asymmetric Information/Signaling</i>		
Proxy: Earnings Growth		-
<i>Pecking Order Theory</i>		
Proxy: Profitability	-	
<i>Maturity Matching</i>		
Proxy: Asset Maturity		+
<i>Market Access</i>		
Proxy: Market Access	+	+

Table 3
Descriptive Statistics

This table presents mean (standard deviation) values for the variables used in this study based on 193 pooled time series cross-sectional observations for REITs over the sample period from 1997 to 2003. Disaggregate evidence by property type is also reported. *Leverage* is the ratio of total debt to the market value of the firm's assets, where the latter is measured as the book value of total assets plus the difference between market value of equity and book value of equity. *Debt Maturity* is the value-weighted debt maturity in months. *Market-to-Book* is the ratio of market value of the firm's assets to book value. Market-to-Book ratio is used as a proxy for growth opportunities. *Firm Size* is market value of the firm's assets in billions of constant 1997 dollars. We use the natural logarithm of size in our regressions. *Market Access* is a dummy variable equal to 1 if the REIT has access to the market for public debt and zero otherwise. We determine access by whether the firm has a public debt rating or not. *Return on Assets* is the ratio of net income to the book value of total assets. *Volatility* is the time series standard deviation of the first differences of funds from operations per share. *Earnings Growth* is the ratio of next year's earnings per share less this year's earnings per share to this year's end share price. *Asset Maturity* is the ratio of real estate investments to the total of depreciation and amortization of real estate investments. *Lease Maturity* is the value-weighted time until lease expiration in months. *Recovery Index* is the weighted average recovery rate calculated using the S&P study of CMBS recoveries by property type and the REIT's property type mix. *Industrial* is a dummy variable equal to 1 if the REIT's investment focus is industrial properties and zero otherwise. Other property type indicators are defined similarly.

Variable	All REITs	Industrial	Apartment	Hotel	Office	Retail
<i>Endogenous Dependent Variables</i>						
Leverage	0.46 (.10)	0.40 (.07)	0.47 (.11)	0.61 (.13)	0.43 (.09)	0.48 (.10)
Debt Maturity (Mo.)	80.43 (34.64)	99.28 (64.00)	98.48 (42.88)	105.10 (40.51)	73.89 (24.03)	72.30 (23.64)
<i>Control Variables</i>						
Market-to-Book	1.16 (.17)	1.29 (.24)	1.16 (.18)	0.91 (.10)	1.11 (.12)	1.20 (.16)
Firm Size (\$B)	3.05 (4.26)	3.21 (2.29)	2.01 (1.73)	1.20 (1.03)	3.73 (5.53)	2.94 (3.96)
Market Access	0.49 (.5)	1.00 (.00)	0.48 (.51)	0.50 (.53)	0.32 (.47)	0.56 (.50)
Return on Assets (%)	3.08 (2.67)	4.32 (1.11)	3.42 (2.91)	-3.91 (5.35)	3.21 (1.85)	3.32 (1.93)
Volatility	0.37 (.51)	0.41 (.19)	0.30 (.17)	1.75 (1.54)	0.34 (.25)	0.26 (.40)
Earnings Growth (%)	17.53 (7.62)	-6.58 (2.26)	-21.89 (3.59)	463.46 (32.63)	-14.06 (4.84)	21.00 (4.59)
Asset Maturity (Yr.)	31.51 (10.17)	31.34 (5.29)	29.21 (8.87)	18.93 (2.57)	32.38 (11.15)	33.08 (10.01)
<i>Proxies for Liquidation Value</i>						
Lease Maturity (Mo.)	58.57 (39.35)	51.08 (8.65)	7.41 (2.33)	0.03 (.00)	66.40 (20.28)	80.76 (43.15)
Industrial	8.29%					
Residential	15.03%					
Hotel	4.15%					
Office	37.31%					
Retail	35.23%					
Number of Observations	193	16	29	16	72	68

Table 4
Two Stage Least Squares Model of Capital Structure Using Traditional Finance Factors Only

This table presents Two-Stage Least Squares regression results using 193 pooled time series cross-sectional observations for REITs over the sample period from 1997 to 2003. *Leverage* is the ratio of total debt to the market value of the firm's assets, where the latter is measured as the book value of total assets plus the difference between market value of equity and book value of equity. *Debt Maturity* is the log of the average debt maturity in months. *Market-to-Book* is the ratio of market value of the firm's assets to book value. Market-to-Book ratio is used as a proxy for growth opportunities. *Firm Size* is market value of the firm's assets. We use the natural log of market value of assets in billions of constant 1997 dollars. *Firm Size Squared* is the square of the natural logarithm of firm size. *Volatility* is the time series standard deviation of the first differences of funds from operations per share. *Earnings Growth* is the ratio of next year's earnings per share less this year's earnings per share to this year's end share price. *Return on Assets* is the ratio of net income to the book value of total assets. *Asset Maturity* is the ratio of real estate investments to the total of depreciation and amortization of real estate investments. *Market Access* is a dummy variable equal to 1 if the REIT has access to the market for public debt and zero otherwise. We determine access by whether the firm has a public debt rating or not.

Variable	Leverage		Debt Maturity	
	Coefficient	t-statistic	Coefficient	t-statistic
<i>Endogenous Variables</i>				
Debt Maturity	-0.2283	-2.00 **	--	--
Leverage	--	--	-1.7685	-1.47
<i>Traditional Control Variables</i>				
Market-to-Book	-0.2659	-3.00 ***	-0.8193	-2.74 ***
ln(Firm Size)	-0.0232	-2.29 ***	-0.6520	-1.88 *
ln(Firm Size) ²	--	--	0.0394	1.76 *
Volatility	0.0411	2.07 **	0.1440	1.78 *
Earnings Growth	--	--	0.1211	0.36
Return on Assets	-0.0088	-2.04 **	--	--
Asset Maturity	--	--	0.0008	0.17
Market Access	0.0168	0.69	0.1042	1.59
Constant	1.9334	3.23 ***	8.5940	3.99 ***
No. of Observations	193		193	
RMSE	0.1085		0.3494	
F Statistic	10.73		3.22	
p-value	0.0000		0.0015	

*** Significant at the 1% level. ** Significant at the 5% level. * Significant at the 10% level.

Table 5
Extended Base Model Using Lease Maturity as Proxy for Liquidation Value

This table presents Two-Stage Least Squares regression results using 193 pooled time series cross-sectional observations for REITs over the sample period from 1997 to 2003. *Leverage* is the ratio of total debt to the market value of the firm's assets, where the latter is measured as the book value of total assets plus the difference between market value of equity and book value of equity. *Debt Maturity* is the log of the average debt maturity in months. *Lease Maturity* is the log of the average lease maturity in months. *Market-to-Book* is the ratio of market value of the firm's assets to book value. Market-to-Book ratio is used as a proxy for growth opportunities. *Firm Size* is the natural log of market value of the firm's assets in billions of constant 1997 dollars. *Firm Size Squared* is the square of the natural logarithm of firm size. *Volatility* is the time series standard deviation of the first differences of funds from operations per share. *Earnings Growth* is the ratio of next year's earnings per share less this year's earnings per share to this year's end share price. *Return on Assets* is the ratio of net income to the book value of total assets. *Asset Maturity* is the ratio of real estate investments to the total of depreciation and amortization of real estate investments. *Market Access* is a dummy variable equal to 1 if the REIT has access to the market for public debt and zero otherwise. We determine access by whether the firm has a public debt rating or not.

Variable	Leverage		Debt Maturity	
	Coefficient	t-statistic	Coefficient	t-statistic
<i>Endogenous Variables</i>				
Debt Maturity	-0.3011	-2.74 ***	--	--
Leverage	--	--	-2.0266	-1.74 *
<i>Liquidation Value Proxy</i>				
Lease Maturity	-0.0252	-1.93 *	-0.0785	-2.72 ***
<i>Traditional Control Variables</i>				
Market-to-Book	-0.2996	-3.41 ***	-0.7950	-2.63 ***
ln(Firm Size)	-0.0170	-1.56	-0.5514	-1.53
ln(Firm Size) ²	--	--	0.0342	1.49
Volatility	0.0322	1.49	0.1055	1.22
Earnings Growth	--	--	0.0766	0.23
Return on Assets	-0.0057	-1.11	--	--
Asset Maturity	--	--	0.0019	0.41
Market Access	0.0144	0.59	0.0670	1.03
Constant	2.3302	3.98 ***	8.5162	3.93 ***
No. of Observations	193		193	
RMSE	0.1231		0.3493	
F Statistic	7.78		4.39	
p-value	0.0000		0.0000	

*** Significant at the 1% level. ** Significant at the 5% level. * Significant at the 10% level.

Table 6
Extended Base Model Using Recovery Index as Proxy for Liquidation Value

This table presents Two-Stage Least Squares regression results using 193 pooled time series cross-sectional observations for REITs over the sample period from 1997 to 2003. *Leverage* is the ratio of total debt to the market value of the firm's assets, where the latter is measured as the book value of total assets plus the difference between market value of equity and book value of equity. *Debt Maturity* is the log of the average debt maturity in months. *Recovery Index* is weighted average recovery rate based on S&P study of CMBS recoveries by property type and the REIT's property type mix. *Market-to-Book* is the ratio of market value of the firm's assets to book value. Market-to-Book ratio is used as a proxy for growth opportunities. *Firm Size* is the natural log of market value of the firm's assets in billions of constant 1997 dollars. *Firm Size Squared* is the square of the natural logarithm of firm size. *Volatility* is the time series standard deviation of the first differences of funds from operations per share. *Earnings Growth* is the ratio of next year's earnings per share less this year's earnings per share to this year's end share price. *Return on Assets* is the ratio of net income to the book value of total assets. *Asset Maturity* is the ratio of real estate investments to the total of depreciation and amortization of real estate investments. *Market Access* is a dummy variable equal to 1 if the REIT has access to the market for public debt and zero otherwise. We determine access by whether the firm has a public debt rating or not.

Variable	Leverage		Debt Maturity	
	Coefficient	t-statistic	Coefficient	t-statistic
<i>Endogenous Variables</i>				
Debt Maturity	-0.2457	-2.16 **	--	--
Leverage	--	--	-1.3377	-1.14
<i>Liquidation Value Proxy</i>				
Recovery Index	0.0042	1.58	0.0154	2.47 **
<i>Traditional Control Variables</i>				
Market-to-Book	-0.2768	-3.09 ***	-0.7509	-2.60 ***
ln(Firm Size)	-0.0226	-2.22 **	-0.5360	-1.59
ln(Firm Size) ²	--	--	0.0323	1.49
Volatility	0.1560	2.25 **	0.1560	2.01 **
Earnings Growth	--	--	0.1844	0.57
Return on Assets	-0.0097	-2.28 **	--	--
Asset Maturity	--	--	0.0023	0.52
Market Access	0.0181	0.74	0.1078	1.71 *
Constant	1.7565	3.45 ***	6.8494	3.12 ***
No. of Observations	193		193	
RMSE	0.1113		0.3356	
F Statistic	8.87		3.79	
p-value	0.0000		0.0001	

*** Significant at the 1% level. ** Significant at the 5% level. * Significant at the 10% level

Table 7
Extended Base Model Using Property Type Indicators as Proxies for Liquidation Value

This table presents Two-Stage Least Squares regression results using 193 pooled time series cross-sectional observations for REITs over the sample period from 1997 to 2003. *Leverage* is the ratio of total debt to the market value of the firm's assets, where the latter is measured as the book value of total assets plus the difference between market value of equity and book value of equity. *Debt Maturity* is the log of the average debt maturity in months. *Industrial* is a dummy variable equal to 1 if the REIT's investment focus is apartment properties and zero otherwise. Other property type indicators are defined similarly. *Market-to-Book* is the ratio of market value of the firm's assets to book value. Market-to-Book ratio is used as a proxy for growth opportunities. *Firm Size* is market value of the firm's assets. We express size in billions of constant 1997 dollars. We use the natural logarithm of size in our regressions. *Firm Size Squared* is the square of the natural logarithm of firm size. *Volatility* is the time series standard deviation of the first differences of funds from operations per share. *Earnings Growth* is the ratio of next year's earnings per share less this year's earnings per share to this year's end share price. *Return on Assets* is the ratio of net income to the book value of total assets. *Asset Maturity* is the ratio of real estate investments to the total of depreciation and amortization of real estate investments. *Market Access* is a dummy variable equal to 1 if the REIT has access to the market for public debt and zero otherwise. We determine access by whether the firm has a public debt rating or not.

Variable	Leverage		Debt Maturity	
	Coefficient	t-statistic	Coefficient	t-statistic
<i>Endogenous Variables</i>				
Debt Maturity	-0.3126	-2.85 ***	--	--
Leverage	--	--	-1.9320	-1.52
<i>Liquidation Proxies</i>				
Industrial	-0.0251	-0.64	-0.0311	-0.28
Hotel	-0.0415	-0.68	-0.0925	-0.50
Office	-0.1364	-3.37 ***	-0.3556	-3.83 ***
Retail	-0.0523	-1.40	-0.1835	-1.87 *
<i>Traditional Control Variables</i>				
Market-to-Book	-0.3715	-3.91 ***	-0.9585	-2.57 ***
ln(Firm Size)	-0.0062	-0.56	-0.5619	-1.57
ln(Firm Size) ²	--	--	0.0366	1.57
Volatility	0.0455	2.02	0.1315	1.28
Earnings Growth	--	--	0.1244	0.39
Return on Assets	-0.0059	-1.19	--	--
Asset Maturity	--	--	0.0011	0.20
Market Access	-0.0186	-0.76	-0.0172	-0.21
Constant	2.3735	4.09 ***	8.5725	3.69 ***
No. of Observations	193		193	
RMSE	0.1204		0.3354	
F Statistic	6.93		4.06	
p-value	0.0000		0.0000	

*** Significant at the 1% level. ** Significant at the 5% level. * Significant at the 10% level